YUNE SANG LEE, Ph.D.

Research Professor · Seoul National University Brain Imaging Center
Email: cogrocker@snu.ac.kr; Website: yunesanglee.com

Research Interests: Cognitive neuroscience of music and language; rhythm-based interventions for neurorehabilitation; digital sound therapy for cognitive and emotional health (binaural beats, immersive audio); aging brain and successful communication

EDUCATION & TRAINING

University of Pennsylvania, Philadelphia, PA — **2011 – 2016**
Department of Neurology & Center for Cognitive Neuroscience
Postdoctoral Research Fellow

Dartmouth College, Hanover, NH — **2005 – 2010**
Department of Psychological and Brain Sciences
Ph.D. in Cognitive Neuroscience

Yonsei University, Seoul, S. Korea — **2001**
Department of System Biology
B.S. in Biology

EMPLOYMENT

Seoul National University — **Oct. 2025 – Present**
Research Professor
SNU Brain Imaging Center / Computational Clinical Science Lab

The University of Texas at Dallas, Richardson, TX — **Aug. 2020 – May 2025**
Assistant Professor
Department of Speech, Language, and Hearing, School of Behavioral and Brain Sciences
Callier Clinical Research Center & Center for BrainHealth

The Ohio State University, Columbus, OH — **Aug. 2016 – July 2020**
Assistant Professor
Department of Speech and Hearing Sciences / Chronic Brain Injury, Discovery Themes

T.K.B Commercial Music Studio, Seoul, S. Korea — **Mar. 2003 – July 2005**
Music Director

Samsung Capital, Seoul, S. Korea — **Jan. 2002 – Feb. 2003**
P.R. Officer

GRANTS

Extramural Funding — Awarded

Development of an AI-based Multidimensional Cognitive Assessment and Personalized Training Prescription Platform for Children with Dyslexia
Agency: MOTIR - Ministry of Trade, Industry, and Resources (산업통상부)
Role: PI | **Total:** KRW 1,190,002,000 | **Duration:** 07/01/2026 – 12/31/2029

Impact of Rigorous Music Training on Brain Development
Agency: Dallas Symphony Orchestra
Role: PI | **Total:** $30,000 | **Duration:** 12/01/2023 – 11/30/2024

TheraBeat: A Novel Sound Therapy for Alzheimer's Patients

Agency: AWARE (awaredallas.org)
Role: PI | **Total:** $40,000 | **Duration:** 05/01/2022 – 03/31/2024

SLAM Lab Research Support
Agency: Neuroscience Innovation Foundation (neurofdn.com)
Role: PI | **Total:** $50,000 | **Duration:** 05/01/2022 – 04/30/2023

Non-invasive Brain Stimulation Using a Novel 3D Sound Therapy
Agency: Digisonic (digisonic.com)
Role: PI | **Total:** $200,000 | **Duration:** 01/01/2022 – 12/31/2022

Speech Hero: A Rhythm-Based Speech Therapy App for Individuals with Aphasia
Agency: NIH Small Business Innovation Research (Grant No: 614362)
Role: Consortium PI | **Total:** $50,000 | **Duration:** 06/01/2020 – 05/31/2021

Investigating the Neural Mechanisms of Language Recovery Through Rhythm Gaming Therapy in Aphasia
Agency: National Institutes of Health — NIDCD (Grant No: 1R21DC018699-01)
Role: PI | **Total:** $429,000 | **Duration:** 01/01/2020 – 12/31/2022 ** No ESI advantage applied*

TRIPODS + X Project: An MBI TGDA + Neuro Program for Undergraduates
Agency: National Science Foundation (Grant No: NSF-CCF 1839356)
Role: co-PI (PI: Janet Best) | **Total:** $200,000 | **Duration:** 09/01/2018 – 08/31/2023 *★ Featured on NSF website front page*

Drum Dance Rehabilitation: A Novel Parkinson's Disease Therapy Program
Agency: The Parkinson's Foundation (Grant No: PF-CGP-19163)
Role: PI | **Total:** $37,000 | **Duration:** 09/01/2019 – 08/31/2020

Application of fNIRS to Neuroscience Research of Speech, Language, and Music
Agency: Shimadzu Scientific Instrument, Japan
Role: PI | **Total:** $380,000 | **Duration:** 12/18/2017 – 12/17/2018

Brain-based Evaluation of SoundMind — a Computerized Cognitive Training Program
Agency: Tech Incubator Program for Startup (TIPS) in Korea (Grant No: S2640193)
Role: PI (sub-contract) | **Total:** $63,000 | **Duration:** 01/01/2019 – 12/31/2019

Music and Language Skills in Korean Children
Agency: Korea Advanced Institute of Science and Technology
Role: co-I (PI: Kyung-Myun Lee) | **Total:** $20,000 | **Duration:** 09/01/2018 – 08/31/2020

Intramural Funding — Awarded

Investigating the Therapeutic Mechanisms of Binaural Beat Stimulation in Children with Developmental Language Disorder
Agency: School of Behavioral and Brain Sciences, University of Texas at Dallas
Role: PI | **Total:** $12,500 | **Duration:** 01/01/2022 – 06/30/2022

Mobile EEG System Request
Agency: School of Behavioral and Brain Sciences, University of Texas at Dallas
Role: co-PI (PI: Maguire) | **Total:** $20,785 | **Duration:** —

Investigating Compensatory Language Processes Prompted by Rhythm Video Game Therapy
Agency: Chronic Brain Injury Discovery Theme, OSU
Role: PI | **Total:** $50,000 | **Duration:** 01/01/2017 – 12/31/2017

A Novel Parkinson's Disease Therapy Program for the Columbus Community
Agency: Connect & Collaborate Grant Program
Role: PI | **Total:** $29,000 | **Duration:** 09/01/2019 – 08/31/2020

OSU Pilot Neuroimaging Grant
Agency: College of Arts and Sciences, OSU
Role: PI | **Total:** $25,000 | **Duration:** 06/01/2018 – 05/31/2019

OSU Social and Behavioral Sciences Small Grant Program
Agency: College of Arts and Sciences, OSU
Role: PI | **Total:** $4,000 | **Duration:** 09/01/2017 – 08/31/2018

OSU Social and Behavioral Sciences Special Grant
Agency: College of Arts and Sciences, OSU
Role: PI | **Total:** $4,000 | **Duration:** 09/22/2018 – 09/21/2019

Travel Awards

ASHA Lesson for Success
Agency: American Speech-Language-Hearing Association
Role: Early-career investigator | **Total:** Full travel + registration | **Duration:** 04/27/2020 – 04/29/2020

The Humanities and the Arts Travel Grant Program
Agency: Global Arts and Humanities, Discovery Themes, OSU
Role: co-I (PI: David Huron) | **Total:** $10,000 | **Duration:** 08/01/2017 – 09/01/2017

Consulting Service

Powerless, Wireless, and Nontoxic Bio-telemetry Sensor and Communication System
Agency: Korea Polytechnic University
Role: Consultant (PI: Jae-Young Chung) | **Total:** — | **Duration:** 02/01/2018 – 01/31/2021

PUBLICATIONS

Selected Publications

Kim, J.H., Kim, H-W., Kovar, J., Lee, Y.S. (2024). Neural consequences of binaural beat stimulation on auditory sentence comprehension: an EEG study. Cerebral Cortex. https://doi.org/10.1093/cercor/bhad459

Kim, H-W., Lee, K., Lee, Y.S. (2023). Sensorimotor and working memory systems jointly support development of perceptual rhythm processing. Developmental Science, 26(1): e13261.

Heard, M.J. & Lee, Y.S. (2020). Shared neural resources of rhythm and grammar: An ALE Meta-Analysis. Neuropsychologia, 137: 107284.

Lee, Y.S., Ahn, S.H., Holt, R.F., Schellenberg, G. (2020). Rhythm and syntax processing in school-age children. Developmental Psychology, 56(9): 1632–1641.

Lee, Y.S., Turkeltaub, P.E., Granger, R.H., Raizada, R.D.S. (2012). Categorical speech processing in Broca's area. Journal of Neuroscience, 32(11): 3942–3948.

Lee, Y.S., Janata, P., Frost, C., Hanke, M., Granger, R.H. (2011). Investigation of melodic contour processing in the brain using pattern-based fMRI. NeuroImage, 57(1): 293–300.

Peer-Reviewed Articles

Henry, M.J., Obleser, J., Crusey, M.R., Fuller, E.R., Lee, Y.S., Meyer, M., et al. (2025). How strong is the rhythm of perception? A registered replication of Hickok et al. (2015). Royal Society Open Science, 12(6), 220497.

Thibodeau, L., Freeman, E., Kronenberger, K., Suarez, E., Kim, H-W., Qi, S., Lee, Y.S. (2025). Exploration of auditory processing beyond speech recognition. Audiology Research, 15(3), 59.

Rashidi, F., Parsaei, M., Kiani, I., Sadri, A., Aarabi, S.R., Lee, Y.S., Moghaddam, H.S. (2024). White matter correlates of impulsive behavior in healthy individuals: A diffusion MRI study. Psychiatry and Clinical Neurosciences Reports, 3(4), e70018.

Schneider, J.M., Kim, J., Poudel, S., Lee, Y.S., Maguire, M. (2024). Environmental experiences and cognitive outcomes are predicted by resting-state EEG in school-aged children. Developmental Cognitive Neuroscience.

Kim, H-W., Park, M.S., Lee, Y.S., Kim, C-Y. (2024). Prior conscious experience modulates the impact of audiovisual temporal correspondence on visual processing outside of awareness. Consciousness and Cognition.

Kim, H-W., Kovar, J., Bajwa, J.S., Miran, Y., Ahmad, A., Moreno, M.M., Price, T.J., Lee, Y.S. (2024). Rhythmic motor behavior explains individual differences in grammar skills in adults. Scientific Reports. https://doi.org/10.1038/s41598-024-53382-9

Kim, J.H., Kim, H-W., Kovar, J., Lee, Y.S. (2024). Neural consequences of binaural beat stimulation on auditory sentence comprehension: an EEG study. Cerebral Cortex. https://doi.org/10.1093/cercor/bhad459

Kim, H-W., Kim, M., McLaren, K., Lee, Y.S. (2024). No influence of regular rhythmic priming on grammaticality judgment and sentence comprehension in English-speaking children. Journal of Experimental Child Psychology, 237, 105760.

Kim, H-W., Happe, J., Lee, Y.S. (2023). Beta and gamma binaural beats enhance auditory sentence comprehension. Psychological Research. https://doi.org/10.1007/s00426-023-01808-w

Kim, H-W., Lee, K., Lee, Y.S. (2023). Sensorimotor and working memory systems jointly support development of perceptual rhythm processing. Developmental Science, 26(1): e13261.

Lee, Y.S., Rogers, C., Min, N.E., Wingfield, A., Grossman, M., Peelle, J.E. (2022). Neural compensation supporting successful speech comprehension in normal aging. Aging Brain, 2: 100051.

Yoon, Y-S., Gutierrez, M., Lee, Y.S. (2021). Effects of the configuration of hearing loss on consonant perception. Journal of American Academy of Audiology, 32(08): 521–527.

Heard, M.J., Li, X., Lee, Y.S. (2021). Hybrid auditory fMRI imaging. Journal of Neuroscience Methods, 358: 109198.

Moritz, M., Heard, M., Kim, H-W., Lee, Y.S. (2020). Invariance of edit-distance to tempo in rhythm similarity. Psychology of Music.

Lee, Y.S., Ahn, S.H., Holt, R.F., Schellenberg, G. (2020). Rhythm and syntax processing in school-age children. Developmental Psychology, 56(9): 1632–1641.

Heard, M.J. & Lee, Y.S. (2020). Shared neural resources of rhythm and grammar: An ALE Meta-Analysis. Neuropsychologia, 137: 107284.

Lee, K.M. & Lee, Y.S. (2019). Beat synchronization development of Korean elementary school students. The Korean Society for Music Theory, 26(2): 251–270.

Lee, Y.S., Wingfield, A., Min, N.E., Kotloff, E., Grossman, M., Peelle, J.E. (2018). Differences in hearing acuity among 'normal-hearing' young adults modulate the neural basis for speech comprehension. eNeuro, 0263-17.2018. ★ Featured in US News & World Report

Lee, Y.S. (2018). Impact of subtle hearing loss on the cognition of young adults. Hearing Journal, 71(10):30. [invited]

Belyk, M., Lee, Y.S., Brown, S. (2018). How does human motor cortex regulate vocal pitch in singers? Royal Society Open Science, 5(8): 172208.

Quandt, L.C., Lee, Y.S., Chatterjee, A. (2017). Neural bases of action abstraction. Biological Psychology, 129: 314–323.

Lee, Y.S., Zreik, J., Hamilton, R.H. (2017). Patterns of neural activity predict picture-naming performance of a patient with chronic aphasia. Neuropsychologia, 94: 52–60.

Lee, Y.S., Min, N.E., Wingfield, A., Grossman, M., Peelle, J.E. (2016). Acoustic richness modulates the neural networks supporting auditory sentence comprehension. Hearing Research, 333: 108–117.

Lee, Y.S., Peelle, J.E., Kraemer, D., Lloyd, S., Granger, R.H. (2015). Multivariate sensitivity to voice during auditory categorization. Journal of Neurophysiology, 114(3): 1819–1826.

Lee, Y.S., Janata, P., Frost, C., Martinez, Z., Granger, R.H. (2015). Melody recognition revisited. Psychonomic Bulletin & Review, 22(1): 163–9.

Wiener, M., Lee, Y.S., Lohoff, F., Coslett, H.B. (2014). Individual differences in morphometry and activation of time perception networks are influenced by genotype. NeuroImage, 89: 10–22.

Raizada, R.D.S. & Lee, Y.S. (2013). Smoothness without smoothing: why Naïve Bayes is not naïve for multisubject searchlight studies. PLOS ONE, 8(7): e69566.

Lee, Y.S., Turkeltaub, P.E., Granger, R.H., Raizada, R.D.S. (2012). Categorical speech processing in Broca's area. Journal of Neuroscience, 32(11): 3942–3948.

Lee, Y.S., Janata, P., Frost, C., Hanke, M., Granger, R.H. (2011). Investigation of melodic contour processing in the brain using pattern-based fMRI. NeuroImage, 57(1): 293–300.

Russ, B.E., Lee, Y.S., Cohen, Y.E. (2007). Neural and behavioral correlates of auditory categorization. Hearing Research, 229: 204–12.

Manuscripts Under Review

Kim, H-W., Kovar, J., Lee, Y.S. From rhythm to language: beta oscillations during tapping predict sentence comprehension.

Wang, L-S, Wang, J-H, Chang, L-T, Lee, Y.S., Kung, C-C. Neural similarity and modularity are not necessarily oppsing properties.

Book Chapters

Lee, Y.S., Wilson, M., Howland, K.M. (2024). Ch 6: Music for speech disorders. In Music Therapy & Music-Based Interventions in Neurology. Springer Nature.

Lee, Y.S., Thaut, C., Santoni, C. (2019). Neurological music therapy for speech and language rehabilitation. Oxford Handbook on Music and the Brain. Oxford University Press.

CONFERENCE PRESENTATIONS (2016 – PRESENT)

29th Annual Meeting of the Korean Society for Brain and Neural Sciences (upcoming) — **Sept. 2026**
From rhythm to language: beta oscillations during tapping predict sentence comprehension
Kim, H-W, Kovar, J., Lee, Y.S. [Poster]

7th Neuroscience Forum for Alzheimer's Disease (NFAD) — **Feb. 2024**
New frontiers of digital sound therapy for neurorehabilitation

Society for Neuroscience (SFN 2023) — **Nov. 2023**
The effect of beta binaural beat stimulation on resting-state brain activity in patients with MCI or AD
Leung, C., Kim, J.H., Happe, J., Kovar, J., Lee, Y.S. [Poster]

Society for Neuroscience (SFN 2023) — **Nov. 2023**
Role of the pre-supplementary motor area in syntax: a tACS study
Leung, C., Heard, M., McLaren, K., Wood, K., Lee, Y.S. [Poster]

Cognitive Neuroscience Society (CNS 2023) — **Mar. 2023**
Neural correlates of enhanced auditory sentence comprehension by binaural beat stimulation
Kim, J.H., Kim, H-W., Kovar, J., Lee, Y.S. [Poster]

Society for Neurobiology of Language (SNL 2022) — **Oct. 2022**
The influence of dopamine genotypes on the relationship between rhythm and language processing
Kim, H-W., Kovar, J., Bajwa, J., Mian, Y., Ahmad, A., Moreno, M.M., Price, T.J., Lee, Y.S. [Poster]

Society for Neurobiology of Language (SNL 2022) — **Oct. 2022**
The effects of rhythm priming on syntactic language processing in children
Kim, H-W., Ginter, K., Lee, Y.S. [Poster]

American Congress of Rehabilitation Medicine (ACRM 2022) — **Nov. 2022**
A novel sound therapy combining music and binaural beats for Alzheimer's disease

Lee, Y.S. [Symposium Talk]

Society for Music Perception & Cognition (SMPC 2022) **Aug. 2022**
Sensorimotor synchronization and auditory working memory jointly predict rhythm perception in children
Kim, H-W., Lee, K.M., Lee, Y.S. [Podium Talk]

Society for Music Perception & Cognition (SMPC 2022) **Aug. 2022**
Prior listening to binaural beats facilitates grammar processing during auditory speech tasks
Happe, J., Kim, H-W., Lee, Y.S. [Poster]

American Congress of Rehabilitation Medicine (ACRM 2019) **Nov. 2019**
Home-based rhythm video gaming therapy for aphasia
Lee, Y.S. [Symposium Talk]

Society for Music Perception and Cognition (SMPC 2019) **Aug. 2019**
Influence of rhythm and beat priming on receptive grammar task
Yen, S., Bendoly, D., Heard, H., Lee, Y.S. [Poster]

Organization of Human Brain Mapping Conference 2019 **June 2019**
Exploring a hybrid auditory fMRI protocol combining ISSS and multiband sequences
Heard, M., Li, X., Wolfe, T., Lee, Y.S. [Poster]

International Conference of Music Perception and Cognition (ICMPC), Montreal **July 2018**
Spontaneous rhythm tapping predicts ability of working memory in children
Lee, Y.S., Corbeil, K., Ahn, S.H. [Symposium Talk]

International Conference of Music Perception and Cognition (ICMPC), Montreal **July 2018**
Shared neural resources of rhythm and grammar: An ALE meta-analysis
Heard, M.J. & Lee, Y.S. [Symposium Talk]

Society for Neurobiology of Language, Baltimore, MD **Nov. 2017**
Differences in hearing acuity among young adults modulate the neural basis for speech communication
Lee, Y.S., Wingfield, A., Min, N.E., Jester, C., Kotloff, E., Grossman, M., Peelle, J.E. [Poster]

Society for Neurobiology of Language, Baltimore, MD **Nov. 2017**
Rhythm sensitivity assists in overcoming acoustic and syntactic challenges during speech listening
Ahn, S., Golthwaite, I., Corbeil, K., Byrer, A., Perry, K., Polakampalli, A., Miller, K., Holt, R.F., Lee, Y.S. [Poster]

INVITED TALKS

2026

The Society for Clinical Music Science Research (upcoming) Oct. 2026
When music Therapy Meets Neuroscience: New Perspectives on Neurorehabilitation

Colloquium, School of Digital Humanities and Computational Social Sciences, KAIST (upcoming) Oct. 2026
Beyond background music: Functional Sound in the Age of Digital Technology

Colloquium, Department of Music Therapy, Hansei University Aug 2026
Music Therapy Meets Neuroscience: New Frontiers in Neurorehabilitation

Korean Music Therapy Association Aug 2026
Music Therapy Meets Neuroscience: New Frontiers in Neurorehabilitation

Music, Mathematics, & Language workshop June 2026
Shared Circuits: How music and language meet in the brain

23rd Avision Biomedical Symposium 2026 June 2026
Functional Sound as Neuromodulation: Turning Brain Rhythms into Mental Regulation

Korean Brain Research Institute (KBRI) May 2026
From Rhythm to Brain Plasticity: Harnessing Music to Transform the Brain

Colloquium, Department of Brain & Cognitive Sciences, KAIST Apr. 2026
Listening to the brain: towards a neuroscience of music for precision mental health

2025

Seminar, College of Electronics and Information Engineering, Kwangwoon University Dec. 2025
New frontiers of digital medicine: the power of functional sound

Seminar, Department of Communication Disorders, Ewha Woman's University Oct. 2025
Shared circuits: how music and language meet in the brain

Colloquium, Department of Music Therapy, Ewha Woman's University Sept. 2025
Beyond metronome: rhythm and the art of intervention

Online Lecture for the Inner Communication and Meditation Aug. 2025
The Brain, Music and Well-being: BMW story

Keynote, Society for Clinical Music Science Research July 2025
The Brain, Music and Well-being: BMW story

1st PKSA (Philadelphia Korean Scholar Association) Korea Workshop — Jan. 2025
The Brain, Music and Well-being

2024

Neuroscience Forum on Alzheimer's Disease, S. Korea — Feb. 2024
New frontiers of digital sound therapy for neurorehabilitation

Science Concert, S. Korea — Feb. 2024
Positive impact of music on brain health

Dallas Symphony Orchestra Guild, Dallas, TX — Jan. 2024
How does musical training impact brain development of school-age children?

2023

CHI Memorial Hospital, Chattanooga, TN — Sept. 2023
Non-invasive brain stimulation using binaural beat combined with music

Vanderbilt University Medical Center, Nashville, TN — Sept. 2023
The Brain, Music, and Wellbeing

Neuroscience Innovation Foundation, St. Louis, MO — Apr. 2023
A novel sound therapy for Alzheimer's disease

The Dallas Symphony Orchestra, Dallas, TX — Feb. 2023
The brain, music, and well-being

2022

Invited Seminar, Seoul National University Hospital, Bundang — Nov. 2022
The brain, music, and well-being

Keynote, AWARE Membership Kick-off Event — Aug. 2022
A novel sound therapy for Alzheimer's disease

Invited Seminar, Laboratory for Cognition and Neurostimulation, University of Pennsylvania — June 2022
The brain, music, and well-being

Special Lecture, Department of Brain & Cognitive Sciences, Seoul National University — May 2022
The brain, music, and well-being

Dartmouth Symposium on Music and Medicine in Cross-Cultural Perspective — Apr. 2022
The brain, music, and well-being

Frontiers Talk Series, Center for BrainHealth — Mar. 2022
The brain, music, and well-being

2021

Center for Vital Longevity Virtual Science Series — Apr. 2021
The brain, music, and well-being

NIH Music & Health Investigator Meeting — Mar. 2021
Investigating the neural mechanisms underlying language recovery through rhythm therapy in aphasia

2020

Therapy Insights (Continuing Education Course) — Dec. 2020
Beat the beat: Rhythm-based intervention for communication disorders

Colloquium, Music and Health Science Research Collaboratory, University of Toronto — Feb. 2020
New frontiers in neurologic music therapy

Colloquium, Cognitive and Behavioral Science, Neuroscience and Music Dept., Washington and Lee University — Feb. 2020
The brain, music, and well-being

2019

Otolaryngology Department Seminar, Washington University in St. Louis, MO — May 2019
The neural and genetic mechanisms underlying the connection between music and language

Topology, Geography, and Data Seminar, OSU — Jan. 2019
The connection between speech, language, and music

2018

Philadelphia Korean Scholar Association (PKSA) Symposium [Keynote] — Dec. 2018
The connection between speech, language, and music

Korea Society for Music Perception and Cognition (KSMPC) [Keynote] — May 2018
The connection between speech, language, and music

Annual Brain Health & Performance Summit, Columbus, OH — Mar. 2018
Rhythm-based rehabilitation for communication disorders

Annual TBI Summit, Columbus, OH — Mar. 2018
The brain, music, and well-being

Cognitive Proseminar, Dept. of Psychology, OSU — Jan. 2018
The connection between speech, language, and music

2017

KSEA SW Ohio Chapter Seminar, University of Cincinnati — Dec. 2017
The brain, music, and well-being

"Aging in Community" International Standardization Symposium, S. Korea — Dec. 2017
Neural compensation supporting successful communication in normal aging

KSEA Ohio Chapter Seminar, The Ohio State University — Apr. 2017
Mind decoding: New frontiers in neuroimaging

2016

Eye and Ear Institute, The Ohio State University — Dec. 2016
Predicting CI outcome using novel neuroimaging techniques

OSU Music Cognition Group, The Ohio State University — Sep. 2016
Rhythm-based rehabilitation for communication disorders

The Haskins Laboratories, New Haven, CT — May 2016
New frontiers in brain-based rehabilitation for communication disorders

AD HOC REVIEWER

American Journal of Speech-Language Pathology · Brain & Language · Cerebral Cortex · Clinical Archives of Communication Disorders · Cortex · European Journal of Neuroscience · Experimental Brain Research · Frontiers in Human Neuroscience · Hearing Research · Human Brain Mapping · International Journal of Psychophysiology · Journal of Alzheimer's Disease · Journal of Cognitive Neuroscience · Journal of Cognitive Science · Journal of Neuroscience · Journal of Neurophysiology · Journal of Neurology · Magnetic Resonance Imaging · Music Perception · NeuroImage · NeuroImage Clinical · Neurocase · Psychomusicology · Scientific Reports

COURSES TAUGHT

Neural Basis of Speech, Language, and Music (undergraduate seminar)	Spring 2021 – 2025
Neural Basis of Speech, Language, and Music (graduate seminar)	Fall 2020 – 2025
Communication, Cognition, and the Aging Brain (undergraduate seminar)	Fall 2023 – 2025
Neurology of Speech and Hearing — SHS 5760, OSU	Fall 2017 – Fall 2019
NeuroImaging for Speech, Language, and Music — SHS 8950, OSU	Spring 2017 – 2019
Statistics — SHS 6786, OSU	Spring 2017 – 2019

Trainees

Matthew Heard, doctoral student	2018 – 2023
Hyun-Woong Kim, doctoral student	2019 – 2024
Carole Leung, doctoral student	2022 – 2024
Saranya Mundayoor, doctoral student	2023 – 2024
Jea-Hong Kim, postdoc	2021 – 2023

UNIVERSITY / COMMUNITY / ACADEMIC SERVICE (2016 – PRESENT)

Director of Scientific Affairs, The Society for Clinical Music Science Research	Jan. 2026 – present
Director of Scientific Affairs, Korean Society for Music Perception & Cognition	Sep. 2016 – present
Callier Postdoc Hiring Committee	Jan. 2024
BBS Committee for Cluster Hire in Lifespan Initiative	Sep. 2022 – 2025
Panelist, National Youth Leadership Forum: Explore STEM Program	June 2022
UTD Intellectual Property Committee	Sep. 2021 – 2025
SLHS Ph.D. Program Admission Committee	Nov. 2021 – 2025
ASHA 2021 Convention Topic Committee	Feb. 2021 – Nov. 2021
BBS Website Renovation Committee	Apr. 2021 – Dec. 2022
BBS Delegate for Blackstone Launchpad Stewardship Council	Aug. 2020 – Aug. 2025
Doctoral Dissertation Committee for Christina Dugas	Aug. 2020 – Aug. 2023
Doctoral Dissertation Committee for Kathryn Kriedler	Dec. 2020 – Feb. 2024
External Reviewer, Swiss National Scientific Foundation	May 2021
External Reviewer, Clinical and Translational Science Awards, Auburn University	Feb. 2020

Session Chair, ACRM Mini-Symposium	Nov. 2019
President, KSEA Ohio Chapter	July 2019 – July 2020
External Reviewer, Clinical and Translational Science Awards, University of Washington	Oct. 2018
Faculty Mentor, OSU STEP (Second-year Transformational Education Program)	Sep. 2017 – Apr. 2019
CBI-Sponsored Monthly fNIRS Seminar Series	Sep. 2017 – 2025
Session Chair, ICMPC	July 2018
Reviewer for STEAM Powered Project Grants	July 2017

MEDIA COVERAGE /Interview

쇼츠 릴스에 빼앗긴 문해력 MBC 뉴스데스크	Mar. 2026
음악을 듣고 연주할 때 뇌에서는 어떤 일이 일어나는가? RYM Collabs.	Mar. 2026
Binaural Beats Boost Language Skills — Neuroscience News	Dec. 2023
Neural Nourishment: How Discovering New Music May Garner Some Serious Brain Regions — Vox	Sep. 2022
Music: A Brain Enhancer? — TwoMaverix	Mar. 2022
NSF: Featuring our fMRI Study Investigating Rhythm and Grammar — NSF Website	Sep. 2018
Podcast Interview with Parsing Neuroscience	Aug. 2018
How Subtle Hearing Loss While Young Changes Brain Function — US News	June 2018
A Minor Hearing Loss Now Can Mean Major Dementia Risk Later — Healthline	July 2018
Early Hearing Loss Could Pave the Way for Dementia — Medical News Today	May 2018
Subtle Hearing Loss While Young Alters Brain Function — Neuroscience News	May 2018